

Finlay Cooper

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Education

Western University, London ON

June 2026

Bachelor of Engineering, Software Engineering | **Dean's Honour List**

- **Core Coursework:** Algorithms & Data Structures; Operating Systems; DBMS; Networking; Web Technologies; Information Security; Machine Learning; Cloud Computing; Software Testing & Maintenance.

Experience

Software Engineering Capstone, GTA Management System | Western University

2025–2026

- Built a full-stack platform (**Next.js / TypeScript**, Node.js / Express, MySQL) **now in active use across Western's Electrical & Computer Engineering department**, replacing a manual spreadsheet process.
- Designed the **configurable matching engine** that replaced manual allocation, scoring every student-course pair on skill match, stated preference, past review ratings, and course history, then packing TA hours at the slot level across **4 seniority-based waves** with cross-term consistency for full-year courses.
- Implemented **non-destructive dry-run assignment**, letting the committee preview results, tune weights, and re-run without ever overwriting human-approved decisions, plus **drag-and-drop** manual reassignment.
- Developed **78 RESTful endpoints** across 14 modules with JWT auth, role- and ownership-based authorization, and Zod validation, exercised end to end by a custom three-role integration harness.
- Built the front end on a **generic, fully-typed CRUD component** reused across 6+ committee screens, plus a browser-side **PDF syllabus importer** that fuzzy-matches extracted topics against the skills catalog.

Software Developer Intern, BroadGrain Commodities | Toronto, ON

Summer 2025

- **Owned the platform end to end as its only engineer**, from requirements through architecture, build, and release; digitization **cut compliance reporting time 75%** across 3 plants, **15%** more from refactoring.
- Drove scoping directly with plant operations managers at each site, translating floor processes and regulatory requirements into system design, then shipped iterative releases into daily production use.
- Consolidated a legacy **.NET** web app and separate **MAUI** mobile app into a single **React Native / TypeScript** codebase shipping to iOS, Android, and web, roughly **10x** the original scope across 11 modules.
- Engineered a **schema-driven backend factory** (Node.js) generating CRUD controllers, RESTful routes, and validation for every entity from one declaration, **collapsing 100+ hand-written files into 3**.
- Built **10 PDF reporting services** (JSReport, Handlebars, digital signatures) on a four-state approval workflow, auto-delivered to **SharePoint via Microsoft Graph + Azure AD** with token caching and backoff.

Projects

Soi Ram | Freelance Restaurant Website | soiram.ca

2025–2026

- Delivered a **production-grade Next.js / TypeScript / Tailwind** site with dynamic theming, fluid `clamp()` typography, and data-driven menu rendering; managed the full client lifecycle from design to deployment.
- Integrated **Toast POS ordering** alongside UberEats and DoorDash for commission-free direct orders.
- Designed a **retro 80s brand identity** with custom typography, mobile-first layout, and SEO optimization.

Fracture Protocol | 2-Person Unity Roguelike

2025–2026

- Architected a **runtime modifier dispatch system** using string IDs queried at runtime, enabling dynamic effect composition across 32 modifiers (5 rarity tiers) without hardcoded branches.
- Implemented **coroutine-driven boss AI** for 3 encounters, running concurrent behaviors (fire patterns, drone phases, respawn timers) with phase transitions at 66%/33% HP; 77 scripts in a 2-person Git workflow.

Skills

Languages & Frameworks: TypeScript/JavaScript, Java, C#, Python, Bash, SQL; React/React Native, Next.js, Node.js, Express.js, Tailwind CSS, Unity

Tools & Platforms: MySQL, PostgreSQL, Firestore, MongoDB; Prisma ORM; Azure AD / SharePoint, Docker, Kubernetes, Nginx, Git, CI/CD, Linux; OAuth 2.0 / OpenID Connect; AI-assisted development (Claude Code); Networking (IP, Routing & Switching)

Spoken Languages: English, French